

MATERIAL SAFETY DATA SHEET(MSDS)

Section 1-Product and Company Information

Product Name

Cypermethrin 35% EC

Company

YINGDE GREATCHEM CHEMICALS CO.,LTD

ADD:Hongfeng District, Shakou Town, Yingde City, Guangdong Province, China P.R.

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Section 2-composition/Information on Ingredient

Substance Name

Cypermethrin

Chemical Name

(RS)-alpha-cyano-3-phenoxybenzyl (1RS)-cis-,
trans-3-(2,2-dichlorovinyl)-2,2-dimethylcyclopropane carboxylate

CAS #

52315-07-8

Formula

C₂₂H₁₉Cl₂NO₃

Composition

Technical 350g/l

Xylene 520g/l

Emulsifier 80g/l

Section 3-Hazards Identification

HAZARDOUS DECOMPOSITION PRODUCTS

May emit toxic fumes on thermal decomposition

HAZARDOUS REACTIONS

Stable under normal temp. & pressure



SENSITIVITY TO JOLTING

None

THERMAL DECOMPOSITION

If involved in a fire, may emit toxic fumes of hydrogen cyanide, chlorine & oxides nitrogen and carbon.

CONDITIONS TO AVOID

Avoid contact with heat, sparks, flames or other sources of ignition.

MATERIALS TO AVOID

Oxidizers

HAZARDOUS DECOMPOSITION PRODUCTS

May emit toxic fumes on thermal decomposition

Very toxic to aquatic organisms. Harmful if swallowed.

Section 4-First Aid Measures

SPECIAL INSTRUCTIONS

INHALATION

Move to fresh air

CONTACT WITH SKIN

Remove all contaminated clothing; wash exposed skin with plenty of soap and water

SPLASHES IN EYES

Flush with plenty of water for at least 15 minutes. Get medical attention if irritation persists

INGESTION

Remove by gastric lavage and catharsis. Maintain blood pressure and airway. Give oxygen if respiration is depressed. Do not perform gastric lavage if victim is unconscious. Get medical attention immediately.

INFORMATION TO DOCTOR OR OTHER TRAINED PERSONS GIVING FIRST AID

No specific antidote known.

Symptomatic treatment should be given.

Section 5-Fire Fighting Measures

SUITABLE EXTINGUISHING MEDIA

Dry chemical, carbon dioxide, halon, water spray or standard foam.



UNSUITABLE EXTINGUISHING MEDIA

Do not scatter spilled material with high pressure water jets

Special exposure hazards in a fire As for all organic materials, combustion may lead to hazardous oxides of carbon and other toxic fumes

SPECIAL PROTECTIVE EQUIP FOR FIRE-FIGHTING

Wear a self-contained respiratory apparatus.

Wear protective clothing .

OTHER INSTRUCTIONS

Cool fire-exposed containers with water-fog & avoid environmental contamination.

Section 6-Accidental Release Measures

PERSONAL PRECAUTIONS

Avoid contact with skin & eyes. No smoking, flames or fires in hazard area.

Keep unnecessary people away.

Isolate hazard area & deny entry.

ENVIRONMENTAL PRECAUTIONS

For larger spills, wet down with water and dike for later disposal.

Disposal must be in accordance with standards applicable to generators of hazardous waste.

METHODS FOR CLEANING UP

Dig holding area for containment.

Use protective cover such as plastic sheet to prevent material from spill in fire extinguisher water.

Use mechanical dredges or lifts to remove immobilized masses of pollution and precipitate.

Section 7-Handling and Storage

HANDLING

SPECIAL PROTECTION MEASURES

When handling chemical do not eat, drink or smoke.. Wash hands and exposed skin before meals and after work.

STORAGE

Sensibility to humidity, light & oxidation

Stable under normal atmospheric conditions when oxidation stored in closed container.

SPECIAL STORAGE REQUIREMENTS



All chemicals must be kept out of reach of children, pets, live stock and people. Pesticides should be stored in a locked, secure place, such as a separate building or storage room.

STORAGE CONDITIONS

Maintain a well ventilated storage area which is relatively free from temperature fluctuations.

Low temperature does not determine a product deterioration but, rather complete solidification. Thus a possible separation or segregation of various isomers.

SHELF LIFE

Will keep their physical & biological properties for minimum 2 years when stored in their original packing in a dry place at an annual average temperature not exceeding 25°C.

RECOMMENDED PACKAGING MATERIAL

PE Lined Mild Steel Drums

SUPPLIER PACKAGING

25/50/200 Kg Steel drums UN-Approved

PACKAGING MATERIAL TO BE AVOIDED

Incompatible with HDPE & plastic containers

INCOMPATIBILITIES

Oxidizers (strong),

Section 8-Exposure Controls/PPE

LIMIT VALUES FOR AIR IN WORKING PLACE

None established

Provide local exhaust or general dilution ventilation system.

PERSONAL PROTECTIVE EQUIPMENT

Respiratory protection	Wear mask with filter for organic vapors
Hand protection	Wear suitable protective gloves
Eye protection	Wear safety glasses or goggles
Skin protection	Wear suitable protective clothing

Section 9-Physical/Chemical Properties



Physical state	Yellow brown viscous liquid/semi solid
Colour and odour	Yellowish brown colour, Mild Odour
pH	0.004 mg KOH/L at 25 °C
Boiling point / boiling range	Decomposes at 220 °C
Melting point	60 - 68 °C
Flash point	>61°C
Flammability (Solid/gas)	Non-flammable
Autoflammability	Non auto flammable
Explosion hazard	Non explosive
Oxidizing properties	Non oxidizing properties
Vapour pressure	1.9 x 10 ⁻⁷ pa (20 °C)
Relative density	1.23 (g/cm ³) at 25 °C
Solubility in water	0.006 mg/L at 25 °C
Solubility in organic solvents:	Acetone - 475 g/l
	Chloroform - 480 g/l
	Xylene - 456 g/l
	Dichloromethane - 501 g/L
	Cyclohexane - 463 g/L
Partition coefficient (Log Pow)	log Pow 5.5 at 20 °C
	log Pow 6.3 at 25 °C

Section 10-Stability and Reactivity

STABILITY AND REACTIVITY

Hazardous reactions	Stable under normal temp. & pressure
Sensitivity to jolting	None
Thermal decomposition	If involved in a fire, may emit toxic fumes of hydrogen cyanide, chlorine & oxides of nitrogen and carbon.
Conditions to avoid	Avoid contact with heat, sparks, flames or other sources of ignition.
Materials to avoid	Oxidizers

Section 11-Toxicological Information



ACUTE TOXICITY

- | | |
|-------------------------------------|-----------------------|
| - LD ₅₀ Oral (rat) | >250 mg/kg bodyweight |
| - LD ₅₀ dermal (rat) | >1600 mg/kg b.w. |
| - LD ₅₀ dermal (rabbit) | > 2000 mg/kg b.w. |
| - LC ₅₀ inhalation (rat) | > 0.7 mg/l air |

CORROSIVENESS**IRRITANCY**

- | | |
|------------------|------------------------------|
| - Eye irritancy | Mild irritant to rabbit eye |
| - Skin irritancy | Mild irritant to rabbit skin |

SENSITIZATION

- | | |
|----------------------|---|
| - Skin sensitization | Mild skin sensitization potential to guinea pigs. |
|----------------------|---|

Section 12- Ecological Information**PERSISTENCE IN THE ENVIRONMENT**

- | | |
|----------------------|--|
| Biodegradation | (Easily) degraded by micro organisms. |
| Chemical degradation | Product breakdown is induced by soil pH.. |
| Mobility | Cypermethrin is immobile in surface waters and when applied to the surface of body of water at rates typical of those used in agriculture application. |

TOXIC EFFECTS ON ORGANISMS

- | | |
|----------------------------------|------------------------|
| Aquatic toxicity | |
| - LC ₅₀ fish | 2-28 g/l (Brown Trout) |
| - LC ₅₀ Daphnia Magna | 0.15 g/litre (48 h) |

OTHER TOXICITY

- | | |
|------------------------------|----------------|
| - LD ₅₀ honey bee | 35 µg a.i./bee |
|------------------------------|----------------|

Section 13-Disposal Considerations**PRODUCTS**

Disposal must be in accordance with standards applicable to generators of hazardous waste

EMPTY PACKAGING

Disposal methods as per state or local codes of practice and laws



Section 14-Transportation Information

LAND TRANSPORT

ADR/RID class

UN No. 3582 Class 9/71C. Pyrethroid Pesticide,
liquid, toxic* (Cypermethrin Tech).

Kemler class

SEA TRANSPORT

- IMDG class

Class 9(Pyrethroid Pesticide, liquid, toxic*
(Cypermethrin Tech) , Severe marine
Pollutant PP)

AIR TRANSPORT

- IATA class

Class 9, (Packing group III) UN No.3582

COMMODITY CODE NO.(H.S. NO.)

29.26.90.99

Section 15-Regulatory Information

EEC NORMS - INFORMATION ON THE WARNING LABEL

Symbol	Xn (Harmful by inhalation and if swallowed and dangerous for the environment)
Risk phrases	R20/22,R 50 , R 53,
Safety advice	S 2, S 24, S 45, S 60.

NATIONAL REGULATIONS

First Aid measures, precautionary statements and storage and disposal instructions to be
quoted on the label.



Section 16-Other Information

MSDS Creation Date:04/18/2005

Revision #0

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